

Week 2 - Assignment 2

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1 NIH Chest X-Ray14 Dataset

- **Type of imaging data:** Chest radiographs (X-ray images) of the thorax.
- **Number of images:** 112,120 frontal chest X-rays from 30,805 patients.
- **Available classes/labels:** 14 disease labels including Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia.
- **Dataset imbalance:** Significant imbalance; common findings like Effusion and Infiltration dominate, while rare conditions such as Hernia have very few examples.
- **Observed challenges:**
 - Labels are text-mined using Natural Language Processing from reports rather than manually annotated, leading to possible label noise.
 - Multi-label nature complicates classification models.
 - Chest X-rays have subtle features that can be hard to distinguish visually.
- **Dataset link:** NIH Chest X-ray14 on Kaggle (Official NIH download: NIH ChestXrayNIHCC)

Summary: The NIH Chest X-ray14 dataset is a large publicly available repository of frontal chest radiographs with multi-label disease annotations. Although extensive in size and variety, the dataset suffers from class imbalance and potential annotation noise due to NLP-based labeling. These factors make it both useful and challenging for research in weakly supervised learning and multi-label classification.

2 HAM10000 (Skin Cancer MNIST) Dataset

- **Type of imaging data:** Dermatoscopic RGB images of pigmented skin lesions.
- **Number of images:** 10,015 dermatoscopic lesion images.

- **Available classes/labels:** Seven lesion categories: Actinic Keratoses, Basal Cell Carcinoma, Benign Keratosis-like Lesions, Dermatofibroma, Melanoma, Melanocytic Nevi, and Vascular Lesions.
- **Dataset imbalance:** Strong imbalance; Melanocytic Nevi dominate the dataset, while classes like Dermatofibroma are rare.
- **Observed challenges:**
 - High visual similarity between certain benign and malignant lesions.
 - Variability in imaging conditions (lighting, skin tone, lesion appearance).
 - Class imbalance risks biasing models toward common classes.
- **Dataset link:** Skin Cancer MNIST: HAM10000 on Kaggle

Summary: The HAM10000 dataset is a benchmark dermatoscopic dataset of seven categories of pigmented skin lesions. Its size and multiple lesion types make it ideal for classification research, but its class imbalance and intra-class variability pose real challenges for robust model development.